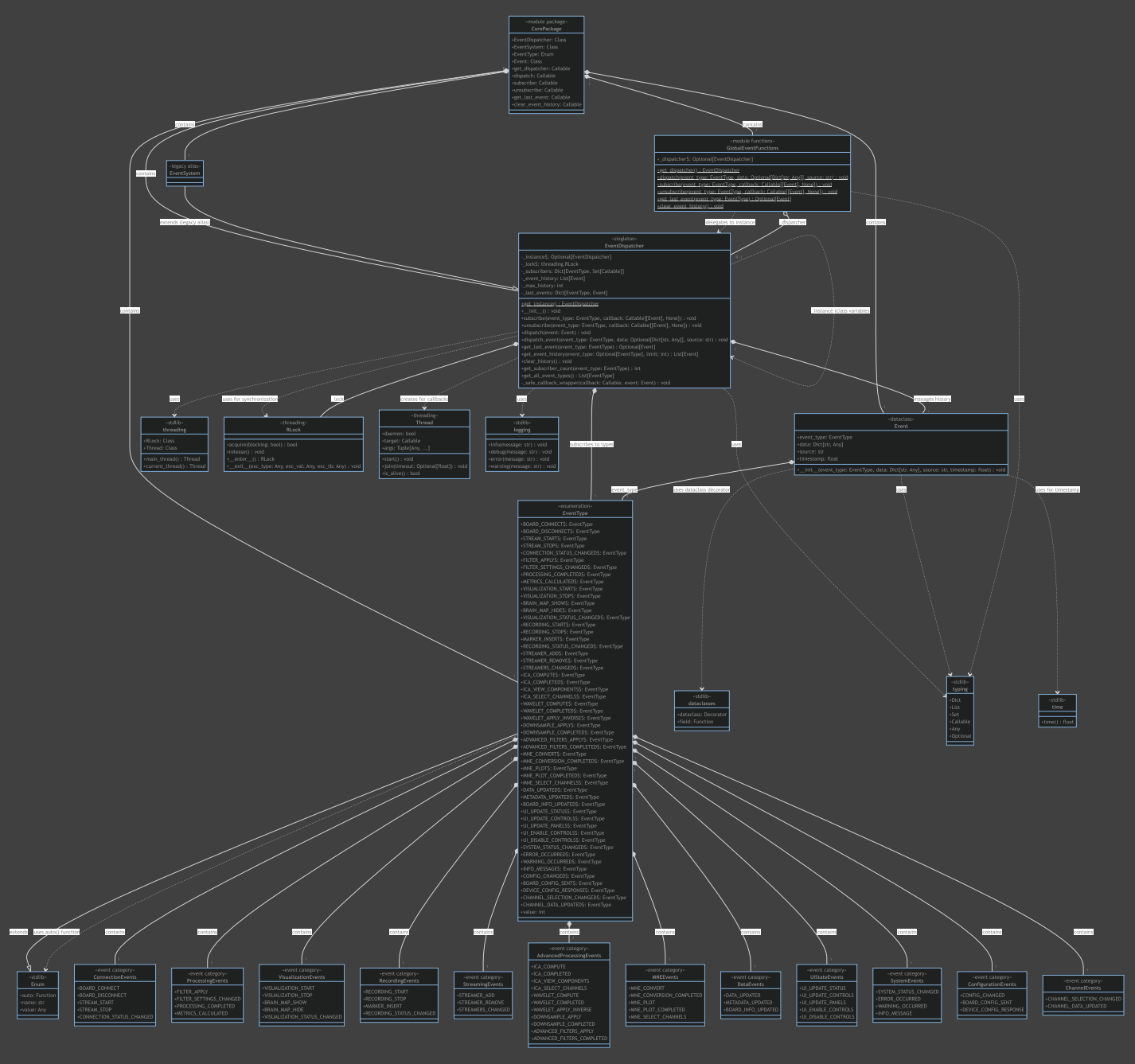

«module package»
CorePackage
+EventDispatcher: Class
+EventSystem: Class
+EventType: Enum
+Event: Class
+get_dispatcher: Callable
+dispatch: Callable
+subscribe: Callable
+unsubscribe: Callable
+get_last_event: Callable
+clear_event_history: Callable
contains
contains
contains
«module functions»
GlobalEventFunctions
- _dispatcher$: Optional[EventDispatcher]
+get_dispatcher() : EventDispatcher
+dispatch(event_type: EventType, data: Optional[Dict[str, Any]], source: str) : void
+subscribe(event_type: EventType, callback: Callable[[Event], None]) : void
+unsubscribe(event_type: EventType, callback: Callable[[Event], None]) : void
+get_last_event(event_type: EventType) : Optional[Event]
+clear_event_history() : void
operate
«legacy alias»
EventSystem
extends (legacy alias)
delegates to instance
dispatches
contains
«singleton»
EventDispatcher
- _instance$: Optional[EventDispatcher]
- _lock$: threading.RLock
- _subscribers: Dict[EventType, Set[Callable]]
- _event_history: List[Event]
- _max_history: int
- _last_events: Dict[EventType, Event]
+get_instance() : EventDispatcher
+ __init__() : void
+subscribe(event_type: EventType, callback: Callable[[Event], None]) : void
+unsubscribe(event_type: EventType, callback: Callable[[Event], None]) : void
+dispatch(event: Event) : void
+dispatch_event(event_type: EventType, data: Optional[Dict[str, Any]], source: str) : void
+get_last_event(event_type: EventType) : Optional[Event]
+get_event_history(event_type: Optional[EventType], limit: int) : List[Event]
+clear_history() : void
+get_subscriber_count(event_type: EventType) : int
+get_all_event_types() : List[EventType]
- _safe_callback_wrapper(callback: Callable, event: Event) : void
instance (class variable)
contains
uses
uses for synchronization
«stdlib»
threading
+RLock: Class
+Thread: Class
+main_thread() : Thread
+current_thread() : Thread
«threading»
RLock
+acquire(blocking: bool) : bool
+release() : void
+__enter__() : RLock
+__exit__(exc_type: Any, exc_val: Any, exc_tb: Any) : void
lock
creates for callback
«threading»
Thread
+daemon: bool
+target: Callable
+args: Tuple[Any, ...]
+start() : void
+join(timeout: Optional[float]) : void
+is_alive() : bool
logs
«stdlib»
logging
+info(message: str) : void
+debug(message: str) : void
+error(message: str) : void
+warning(message: str) : void
manages history
«dataclass»
Event
+event_type: EventType
+data: Dict[str, Any]
+source: str
+timestamp: float
+ __init__(event_type: EventType, data: Dict[str, Any], source: str, timestamp: float) : void
uses
uses for timestamp
«stdlib»
time
+time() : float
subscribes to types
«enumeration»
EventType
+BOARD_CONNECTS: EventType
+BOARD_DISCONNECTS: EventType
+STREAM_STARTS: EventType
+STREAM_STOPS: EventType
+CONNECTION_STATUS_CHANGEDS: EventType
+FILTER_APPLYS: EventType
+FILTER_SETTINGS_CHANGEDS: EventType
+PROCESSING_COMPLETEDS: EventType
+METRICS_CALCULATEDS: EventType
+VISUALIZATION_STARTS: EventType
+VISUALIZATION_STOPS: EventType
+BRAIN_MAP_SHOWS: EventType
+BRAIN_MAP_HIDES: EventType
+VISUALIZATION_STATUS_CHANGEDS: EventType
+RECORDING_STARTS: EventType
+RECORDING_STOPS: EventType
+MARKER_INSERTS: EventType
+RECORDING_STATUS_CHANGEDS: EventType
+STREAMER_ADDS: EventType
+STREAMER_REMOVES: EventType
+STREAMERS_CHANGEDS: EventType
+ICA_COMPUTES: EventType
+ICA_COMPLETEDS: EventType
+ICA_VIEW_COMPONENTS: EventType
+ICA_SELECT_CHANNELSS: EventType
+WAVELET_COMPUTES: EventType
+WAVELET_COMPLETEDS: EventType
+WAVELET_APPLY_INVERSES: EventType
+DOWNSAMPLE_APPLYS: EventType
+DOWNSAMPLE_COMPLETEDS: EventType
+ADVANCED_FILTERS_APPLYS: EventType
+ADVANCED_FILTERS_COMPLETEDS: EventType
+MNE_CONVERTS: EventType
+MNE_CONVERSION_COMPLETEDS: EventType
+MNE_PLOTS: EventType
+MNE_PLOT_COMPLETEDS: EventType
+MNE_SELECT_CHANNELSS: EventType
+DATA_UPDATEDS: EventType
+METADATA_UPDATEDS: EventType
+BOARD_INFO_UPDATEDS: EventType
+UI_UPDATE_STATUSS: EventType
+UI_UPDATE_CONTROLSS: EventType
+UI_UPDATE_PANELSS: EventType
+UI_ENABLE_CONTROLSS: EventType
+UI_DISABLE_CONTROLSS: EventType
+SYSTEM_STATUS_CHANGEDS: EventType
+ERROR_OCCURREDS: EventType
+WARNING_OCCURREDS: EventType
+INFO_MESSAGES: EventType
+CONFIG_CHANGEDS: EventType
+BOARD_CONFIG_SENTS: EventType
+DEVICE_CONFIG_RESPONSES: EventType
+CHANNEL_SELECTION_CHANGEDS: EventType
+CHANNEL_DATA_UPDATEDS: EventType
+value: int
event_type
uses dataclass decorator
«stdlib»
dataclasses
+dataclass: Decorator
+field: Function
«stdlib»
typing
+Dict
+List
+Set
+Callable
+Any
+Optional
extends
uses auto() function
«stdlib»
Enum
+auto: Function
+name: str
+value: Any
contains
«event category»
ConnectionEvents
+BOARD_CONNECT
+BOARD_DISCONNECT
+STREAM_START
+STREAM_STOP
+CONNECTION_STATUS_CHANGED
contains
«event category»
ProcessingEvents
+FILTER_APPLY
+FILTER_SETTINGS_CHANGED
+PROCESSING_COMPLETED
+METRICS_CALCULATED
contains
«event category»
VisualizationEvents
+VISUALIZATION_START
+VISUALIZATION_STOP
+BRAIN_MAP_SHOW
+BRAIN_MAP_HIDE
+VISUALIZATION_STATUS_CHANGED
contains
«event category»
RecordingEvents
+RECORDING_START
+RECORDING_STOP
+MARKER_INSERT
+RECORDING_STATUS_CHANGED
contains
«event category»
StreamingEvents
+STREAMER_ADD
+STREAMER_REMOVE
+STREAMERS_CHANGED
contains
«event category»
AdvancedProcessingEvents
+ICA_COMPUTE
+ICA_COMPLETED
+ICA_VIEW_COMPONENTS
+ICA_SELECT_CHANNELS
+WAVELET_COMPUTE
+WAVELET_COMPLETED
+WAVELET_APPLY_INVERSE
+DOWNSAMPLE_APPLY
+DOWNSAMPLE_COMPLETED
+ADVANCED_FILTERS_APPLY
+ADVANCED_FILTERS_COMPLETED
contains
«event category»
MNEEvents
+MNE_CONVERT
+MNE_CONVERSION_COMPLETED
+MNE_PLOT
+MNE_PLOT_COMPLETED
+MNE_SELECT_CHANNELS
contains
«event category»
DataEvents
+DATA_UPDATED
+METADATA_UPDATED
+BOARD_INFO_UPDATED
contains
«event category»
UIStateEvents
+UI_UPDATE_STATUS
+UI_UPDATE_CONTROLS
+UI_UPDATE_PANELS
+UI_ENABLE_CONTROLS
+UI_DISABLE_CONTROLS
contains
«event category»
SystemEvents
+SYSTEM_STATUS_CHANGED
+ERROR_OCCURRED
+WARNING_OCCURRED
+INFO_MESSAGE
contains
«event category»
ConfigurationEvents
+CONFIG_CHANGED
+BOARD_CONFIG_SENT
+DEVICE_CONFIG_RESPONSE
contains
«event category»
ChannelEvents
+CHANNEL_SELECTION_CHANGED
+CHANNEL_DATA_UPDATED